# DBMS Assignment-3

## Topic : Hostel Management System

### Member Details

1)Saicharan Polishetty-PES1UG19CS329
2)Pavan Karthick-PES1UG19CS325
3)P C Sai Preetam-PES1UG19CS314

### Simple Queries

1)Retrive all the relations

```
postgres=# \c hostel_management
You are now connected to database "hostel_management" as user "postgres".
hostel_management=# \d
              List of relations
 Schema |       Name        | Type  |  Owner
--------+-------------------+-------+----------
 public | blocks            | table | postgres
 public | do_laundry        | table | postgres
 public | eats_in           | table | postgres
 public | facilitates       | table | postgres
 public | fee               | table | postgres
 public | goes_for          | table | postgres
 public | has               | table | postgres
 public | hostel            | table | postgres
 public | laundry           | table | postgres
 public | mess              | table | postgres
 public | outing            | table | postgres
 public | pays              | table | postgres
 public | room              | table | postgres
 public | staff             | table | postgres
 public | staff_mail_id     | table | postgres
 public | staff_phone_no    | table | postgres
 public | student           | table | postgres
 public | student_mail_id   | table | postgres
 public | student_phone_no  | table | postgres
 public | visitor           | table | postgres
 public | visits            | table | postgres
 public | works_in          | table | postgres
(22 rows)
```

2)Retrive all the Information from student

```
hostel_management=# select * from student;
  srn   |     name      |    dob     | cgpa | year_of_study |       address
--------+---------------+------------+------+---------------+---------------------
 CS329  | Saicharan     | 2002-02-19 | 7.81 |          2021 | Hyderabad Telangana
 CS325  | Pavan Karthik | 2002-05-30 | 7.76 |          2021 | Bengaluru
 CS314  | Preetam       | 2001-09-11 | 7.55 |          2022 | Hindupur
 EC908  | Krishna       | 2000-10-18 | 9.95 |          2020 | Kurnool Andhra
 CV015  | Balaram       | 1999-07-25 | 8.77 |          2019 | Mysore Karnataka
 EEE584 | Nikhil        | 1990-10-19 |    9 |          2011 | Dubai
(6 rows)
```

3)Retrieve all the Names from Staff

```
hostel_management=# select Staff_Name from staff;
 staff_name
------------
 Kishore
 Mendes
 Dhoni
 Arjun
 Karan
 Cristiano
 Benzema
 Jagan
 Weeknd
 Eminem
(10 rows)
```

4)Retrieve all the Transaction-ID's from Fee

```
hostel_management=# select Transaction_Id from Fee;
 transaction_id
----------------
 123456786
 123456789
 123456785
 123456784
 123456783
 123456782
(6 rows)
```

5)Inserting a Sudent's record in the Database

```
hostel_management=# INSERT INTO Student Values('BT123','Rahul','2000-08-09',9.57,2020,'Dubai');
INSERT 0 1
hostel_management=# select Name from Student;
     name
--------------
 Saicharan
 Pavan Karthik
 Preetam
 Krishna
 Balaram
 Nikhil
 Rahul
(7 rows)
```

```
hostel_management=# select * from Student;
  srn   |     name      |    dob     | cgpa | year_of_study |       address
--------+---------------+------------+------+---------------+---------------------
 CS329  | Saicharan     | 2002-02-19 | 7.81 |          2021 | Hyderabad Telangana
 CS325  | Pavan Karthik | 2002-05-30 | 7.76 |          2021 | Bengaluru
 CS314  | Preetam       | 2001-09-11 | 7.55 |          2022 | Hindupur
 EC908  | Krishna       | 2000-10-18 | 9.95 |          2020 | Kurnool Andhra
 CV015  | Balaram       | 1999-07-25 | 8.77 |          2019 | Mysore Karnataka
 EEE584 | Nikhil        | 1990-10-19 |    9 |          2011 | Dubai
 BT123  | Rahul         | 2000-08-09 | 9.57 |          2020 | Dubai
(7 rows)
```

6)Retrive all the Student-Mail ID's

```
hostel_management=# select Mail_Id from Student_Mail_Id;
          mail_id
---------------------------
 saicharan333@gmail.com
 polisaicharan420@gmail.com
 mpkarthick2002@gmail.com
 saipreetham@gmail.com
 balakrishna@gmail.com
 balramthor@gmail.com
 nikhil69@gmail.com
(7 rows)
```

## Complex Queries

1)Retrive all the Names from Student that aren't in Outing List

```
hostel_management=# select Name from Student where not exists(select * from Outing where SRN=Out_SRN);
 name
-------
 Rahul
(1 row)
```

2)Maximum price of Fee in the database

```
hostel_management=# select * from Fee where Amount=(select max(Amount) from Fee);
 transaction_id | payment_type | remitter_name | amount |    date
----------------+--------------+---------------+--------+------------
 123456782      | IMPS         | Shawn         | 520000 | 09-08-2021
(1 row)
```

3)Count of Students coming from different places

```
postgres=# \c hostel_management
You are now connected to database "hostel_management" as user "postgres".
hostel_management=# select Address as place,count(*) from Student group by Address;
   place    | count
-----------+-------
 Hindupur  |     3
 Bengaluru |     1
 Hyderabad |     2
(3 rows)
```

4)Retrive Names of Staff who works in Mess and who gives Outing, Retrive Names of Staff who works in Food Court and gives Outing

```
hostel_management=# (select Staff_Name from Works_In) INTERSECT (select Staff_Name from Facilitates);
 staff_name
------------
 Eminem
 Kishore
 Dhoni
(3 rows)

hostel_management=# (select Staff_Name from Works_In where Mess_Name='Food Court') INTERSECT (select Staff_Name from Facilitates);
 staff_name
------------
 Dhoni
(1 row)
```

5)Retrive Names from Staff who's not working in Mess and Outing

```
hostel_management=# (select Staff_Name from Staff) EXCEPT ((select Staff_Name from Works_In) UNION (select Staff_Name from Facilitates));
 staff_name
-----------
 Mendes
 Benzema
(2 rows)
```

6)Retrive SRN of students who paid the Fee where Payment Type='NEFT'

```
hostel_management=# select p.SRN as Id,e.Payment_Type as event from Pays as p,Fee as e where p.Transaction_id=e.Transaction_Id AND e.Payment_Type='NEFT';
  id   | event
-------+-------
 CS329 | NEFT
 CS314 | NEFT
(2 rows)
```

## Multiple users with different access privilege levels for different parts of the Database

1)Create user 'Administrator' and grant privileges on the table

```
hostel_management=# create user Administrator with password 'admin' createdb;
CREATE ROLE
hostel_management=# grant all privileges on all tables in schema public to Administrator;
GRANT
hostel_management=# _
```

```
saic@saic:~$ psql -h localhost -d hostel_management -U administrator -p 5432 -W
Password:
psql (12.8 (Ubuntu 12.8-0ubuntu0.20.04.1))
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, bits: 256, compression: off)
Type "help" for help.

hostel_management=> select * from student;
  srn   |     name      |    dob     | cgpa | year_of_study |  address
--------+---------------+------------+------+---------------+-----------
 CS329  | Saicharan     | 2002-02-19 | 7.81 |          2021 | Hyderabad
 CS325  | Pavan Karthik | 2002-05-30 | 7.76 |          2021 | Bengaluru
 CS314  | Preetam       | 2001-09-11 | 7.55 |          2022 | Hindupur
 EC908  | Krishna       | 2000-10-18 | 9.95 |          2020 | Hindupur
 CV015  | Balaram       | 1999-07-25 | 8.77 |          2019 | Hindupur
 EEE584 | Nikhil        | 1990-10-19 |    9 |          2011 | Hyderabad
(6 rows)

hostel_management=> INSERT INTO Student values('ME123','Shyam','1989-02-23',9.75,2020,'Delhi');
INSERT 0 1
hostel_management=> select * from student;
  srn   |     name      |    dob     | cgpa | year_of_study |  address
--------+---------------+------------+------+---------------+-----------
 CS329  | Saicharan     | 2002-02-19 | 7.81 |          2021 | Hyderabad
 CS325  | Pavan Karthik | 2002-05-30 | 7.76 |          2021 | Bengaluru
 CS314  | Preetam       | 2001-09-11 | 7.55 |          2022 | Hindupur
 EC908  | Krishna       | 2000-10-18 | 9.95 |          2020 | Hindupur
 CV015  | Balaram       | 1999-07-25 | 8.77 |          2019 | Hindupur
 EEE584 | Nikhil        | 1990-10-19 |    9 |          2011 | Hyderabad
 ME123  | Shyam         | 1989-02-23 | 9.75 |          2020 | Delhi
(7 rows)

hostel_management=> _
```

2)Creating roles for customer who can view Information of Students,Visitors,Satff

```
hostel_management=# create user charan with password 'charan' createdb;
CREATE ROLE
hostel_management=# create user pavan with password 'pavan' createdb;
CREATE ROLE
hostel_management=# create user preetam with password 'preetam' createdb;
CREATE ROLE
```

```
hostel_management=# grant select on Student to charan;
GRANT
hostel_management=# grant select on Visitor to pavan;
GRANT
hostel_management=# grant select on Staff to preetam;
GRANT
hostel_management=# _
```

```
saic@saic:~$ psql -h localhost -d hostel_management -U charan -p 5432 -W
Password:
psql (12.8 (Ubuntu 12.8-0ubuntu0.20.04.1))
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, bits: 256, compression: off)
Type "help" for help.

hostel_management=> select * from Student;
  srn   |     name     |    dob     | cgpa | year_of_study |  address
--------+--------------+------------+------+---------------+-----------
 CS329  | Saicharan    | 2002-02-19 | 7.81 |          2021 | Hyderabad
 CS325  | Pavan Karthik| 2002-05-30 | 7.76 |          2021 | Bengaluru
 CS314  | Preetam      | 2001-09-11 | 7.55 |          2022 | Hindupur
 EC908  | Krishna      | 2000-10-18 | 9.95 |          2020 | Hindupur
 CV015  | Balaram      | 1999-07-25 | 8.77 |          2019 | Hindupur
 EEE584 | Nikhil       | 1990-10-19 |    9 |          2011 | Hyderabad
(6 rows)

hostel_management=> _
```

```
saic@saic:~$ psql -h localhost -d hostel_management -U pavan -p 5432 -W
Password:
psql (12.8 (Ubuntu 12.8-0ubuntu0.20.04.1))
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, bits: 256, compression: off)
Type "help" for help.

hostel_management=> select * from Visitor;
 visitor_name | out_time | in_time  |    date    | related_as
--------------+----------+----------+------------+------------
 Sreekar      | 17:30:00 | 07:30:00 | 2021-10-27 | Friend
 Kundan       | 19:30:00 | 05:45:00 | 2021-07-12 | Friend
 Thushar      | 20:15:00 | 09:00:00 | 2020-12-13 | Brother
 Sridhar      | 16:15:00 | 08:00:00 | 1999-1-5   | Father
 Rakesh       | 15:45:00 | 11:45:00 | 1999-07-22 | Brother
 Varshith     | 19:20:00 | 09:20:00 | 2020-11-29 | Brother
(6 rows)
```

```
saic@saic:~$ psql -h localhost -d hostel_management -U preetam -p 5432 -W
Password:
psql (12.8 (Ubuntu 12.8-0ubuntu0.20.04.1))
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, bits: 256, compression: off)
Type "help" for help.

hostel_management=> select * from Staff;
 staff_name |       address
------------+---------------------
 Kishore    | Bengaluru Karnataka
 Mendes     | Vizag AndhraPradesh
 Dhoni      | Chennai
 Arjun      | Kochi
 Karan      | Mumbai
 Cristiano  | Amaravati
 Benzema    | Los Angeles
 Jagan      | Ahmedabad
 Weeknd     | Maldives
 Eminem     | Brooklyn
(10 rows)

hostel_management=> _
```

## Transaction

1)Create a transaction consisting of a create table and multiple insert statements.After End transaction the changes should be committed and can be checked using select statement.

```
postgres=# \c hostel_management
You are now connected to database "hostel_management" as user "postgres".
hostel_management=# begin;
BEGIN
hostel_management=# create table table1(Name varchar(30), SRN varchar(15),Phone_No Bigint);
CREATE TABLE
hostel_management=# insert into table1 values('Sreekar','CS065',6304667241);
INSERT 0 1
hostel_management=# insert into table1 values('Hema','EEE123',9959418352);
INSERT 0 1
hostel_management=# insert into table1 values('Rohan','ECE250',9291462555);
INSERT 0 1
hostel_management=# select * from table1;
  name   |  srn   |  phone_no
---------+--------+------------
 Sreekar | CS065  | 6304667241
 Hema    | EEE123 | 9959418352
 Rohan   | ECE250 | 9291462555
(3 rows)

hostel_management=# end transaction;
COMMIT
hostel_management=# select * from table1;
  name   |  srn   |  phone_no
---------+--------+------------
 Sreekar | CS065  | 6304667241
 Hema    | EEE123 | 9959418352
 Rohan   | ECE250 | 9291462555
(3 rows)
```

2)For the above transaction introduce a roll back after inserting 2 records. The create and insert should not be reflected in the database.

```
hostel_management=# begin;
BEGIN
hostel_management=# create table table3(Name varchar(30), SRN varchar(15),Phone_No Bigint);
CREATE TABLE
hostel_management=# insert into table3 values('Priya','CSE124',7032814690);
INSERT 0 1
hostel_management=# insert into table3 values('Pranav','EEE190',7093448441);
INSERT 0 1
hostel_management=# rollback;
ROLLBACK
hostel_management=# select * from table3;
ERROR:  relation "table3" does not exist
LINE 1: select * from table3;
                      ^
```

3)For the first transaction introduce a save point after inserting 2 records and insert 2 more records and rollback to savepoint.They database should reflect only first 2 insert.

```
hostel_management=# begin;
BEGIN
hostel_management=# create table table10(Name varchar(30), SRN varchar(15),Phone_No Bigint);
CREATE TABLE
hostel_management=# insert into table10 values('Akash','ECE012',99999999);
INSERT 0 1
hostel_management=# insert into table10 values('Raga','CV011',8888888888);
INSERT 0 1
hostel_management=# savepoint s1;
SAVEPOINT
hostel_management=# insert into table10 values('Shiva','ECE145',9121713302);
INSERT 0 1
hostel_management=# insert into table10 values('Hari','CV945',8106490290);
INSERT 0 1
hostel_management=# select * from table10;
 name  |  srn   |  phone_no
-------+--------+------------
 Akash | ECE012 |   99999999
 Raga  | CV011  | 8888888888
 Shiva | ECE145 | 9121713302
 Hari  | CV945  | 8106490290
(4 rows)
```

```
hostel_management=# rollback to s1;
ROLLBACK
hostel_management=# select * from table10;
 name  |  srn   |  phone_no
-------+--------+------------
 Akash | ECE012 |   99999999
 Raga  | CV011  | 8888888888
(2 rows)

hostel_management=# _
```

## Concurrenct Transaction

**To display concurrent transactions, we use 2 command prompts.**

1)Here, We try to insert new record of Student and Staff,at the same instant concurrently we will try to see the result (like if the records are added).

Here we begin transaction without committing: (1'st cmd)

```
postgres=# \c hostel_management
You are now connected to database "hostel_management" as user "postgres".
hostel_management=# begin;
BEGIN
hostel_management=# insert into Student values('ME123','Rahul','2001-04-21',8.64,2017,'Mumbai');
INSERT 0 1
hostel_management=# insert into Staff values('Rohan','Delhi');
INSERT 0 1
```

Here we show if the result is updated: ( 2'nd cmd) (It doesn’t since transaction isn’t committed):

```
postgres=# \c hostel_management
You are now connected to database "hostel_management" as user "postgres".
hostel_management=# select * from student;
  srn   |     name      |    dob     | cgpa | year_of_study |  address
--------+---------------+------------+------+---------------+-----------
 CS329  | Saicharan     | 2002-02-19 | 7.81 |          2021 | Hyderabad
 CS325  | Pavan Karthik | 2002-05-30 | 7.76 |          2021 | Bengaluru
 CS314  | Preetam       | 2001-09-11 | 7.55 |          2022 | Hindupur
 EC908  | Krishna       | 2000-10-18 | 9.95 |          2020 | Hindupur
 CV015  | Balaram       | 1999-07-25 | 8.77 |          2019 | Hindupur
 EEE584 | Nikhil        | 1990-10-19 |    9 |          2011 | Hyderabad
(6 rows)

hostel_management=# select * from Staff;
 staff_name |       address
------------+--------------------
 Kishore    | Bengaluru Karnataka
 Mendes     | Vizag AndhraPradesh
 Dhoni      | Chennai
 Arjun      | Kochi
 Karan      | Mumbai
 Cristiano  | Amaravati
 Benzema    | Los Angeles
 Jagan      | Ahmedabad
 Weeknd     | Maldives
 Eminem     | Brooklyn
(10 rows)
```

Here we commit the transaction: (1'st cmd)

```
postgres=# \c hostel_management
You are now connected to database "hostel_management" as user "postgres".
hostel_management=# begin;
BEGIN
hostel_management=# insert into Student values('ME123','Rahul','2001-04-21',8.64,2017,'Mumbai');
INSERT 0 1
hostel_management=# insert into Staff values('Rohan','Delhi');
INSERT 0 1
hostel_management=# END;
COMMIT
hostel_management=# _
```

And show if updated at 2'nd cmd : ( we see the records are added )

```
hostel_management=# select * from student;
  srn   |     name      |    dob     | cgpa | year_of_study |  address
--------+---------------+------------+------+---------------+-----------
 CS329  | Saicharan     | 2002-02-19 | 7.81 |          2021 | Hyderabad
 CS325  | Pavan Karthik | 2002-05-30 | 7.76 |          2021 | Bengaluru
 CS314  | Preetam       | 2001-09-11 | 7.55 |          2022 | Hindupur
 EC908  | Krishna       | 2000-10-18 | 9.95 |          2020 | Hindupur
 CV015  | Balaram       | 1999-07-25 | 8.77 |          2019 | Hindupur
 EEE584 | Nikhil        | 1990-10-19 |    9 |          2011 | Hyderabad
 ME123  | Rahul         | 2001-04-21 | 8.64 |          2017 | Mumbai
(7 rows)

hostel_management=# select * from Staff;
 staff_name |       address
------------+--------------------
 Kishore    | Bengaluru Karnataka
 Mendes     | Vizag AndhraPradesh
 Dhoni      | Chennai
 Arjun      | Kochi
 Karan      | Mumbai
 Cristiano  | Amaravati
 Benzema    | Los Angeles
 Jagan      | Ahmedabad
 Weeknd     | Maldives
 Eminem     | Brooklyn
 Rohan      | Delhi
(11 rows)

hostel_management=# _
```

2)Here we try to update the Address of Student from 'Hindupur' to 'Dubai'

Here we begin transaction in (1'st cmd) and showing in (2'nd cmd):

```
hostel_management=# begin;
BEGIN
hostel_management=# update Student set Address='Dubai' where Address='Hindupur';
UPDATE 3
```

Here in 2'nd cmd we show if the Address is updated (not updated since transaction in not ended):

```
hostel_management=# select * from Student;
  srn   |     name      |    dob     | cgpa | year_of_study |  address
--------+---------------+------------+------+---------------+-----------
 CS329  | Saicharan     | 2002-02-19 | 7.81 |          2021 | Hyderabad
 CS325  | Pavan Karthik | 2002-05-30 | 7.76 |          2021 | Bengaluru
 CS314  | Preetam       | 2001-09-11 | 7.55 |          2022 | Hindupur
 EC908  | Krishna       | 2000-10-18 | 9.95 |          2020 | Hindupur
 CV015  | Balaram       | 1999-07-25 | 8.77 |          2019 | Hindupur
 EEE584 | Nikhil        | 1990-10-19 |    9 |          2011 | Hyderabad
 ME123  | Rahul         | 2001-04-21 | 8.64 |          2017 | Mumbai
(7 rows)
```

Here we end the update transaction: (1'st cmd)

```
hostel_management=# begin;
BEGIN
hostel_management=# update Student set Address='Dubai' where Address='Hindupur';
UPDATE 3
hostel_management=# end;
COMMIT
hostel_management=# _
```

Here in 2'nd cmd we show the updated Address

```
hostel_management=# select * from Student;
  srn   |     name      |    dob     | cgpa | year_of_study |  address
--------+---------------+------------+------+---------------+-----------
 CS329  | Saicharan     | 2002-02-19 | 7.81 |          2021 | Hyderabad
 CS325  | Pavan Karthik | 2002-05-30 | 7.76 |          2021 | Bengaluru
 EEE584 | Nikhil        | 1990-10-19 |    9 |          2011 | Hyderabad
 ME123  | Rahul         | 2001-04-21 | 8.64 |          2017 | Mumbai
 CS314  | Preetam       | 2001-09-11 | 7.55 |          2022 | Dubai
 EC908  | Krishna       | 2000-10-18 | 9.95 |          2020 | Dubai
 CV015  | Balaram       | 1999-07-25 | 8.77 |          2019 | Dubai
(7 rows)

hostel_management=# _
```

## Performance Analysis

```
postgres=# \c hostel_management
You are now connected to database "hostel_management" as user "postgres".
hostel_management=# EXPLAIN ANALYZE select Name,Address from Student where Address NOT IN (select Address from Student where Address='Hyderabad');
```

```
                                                   QUERY PLAN
-------------------------------------------------------------------------------------------------------------
 Seq Scan on student  (cost=16.00..32.00 rows=240 width=64) (actual time=0.072..0.077 rows=4 loops=1)
   Filter: (NOT (hashed SubPlan 1))
   Rows Removed by Filter: 2
   SubPlan 1
     ->  Seq Scan on student student_1  (cost=0.00..16.00 rows=2 width=32) (actual time=0.010..0.014 rows=2 loops=1)
           Filter: ((address)::text = 'Hyderabad'::text)
           Rows Removed by Filter: 4
 Planning Time: 0.662 ms
 Execution Time: 0.258 ms
(9 rows)

~
~
~
~
```

```
postgres=# \c hostel_management
You are now connected to database "hostel_management" as user "postgres".
hostel_management=# EXPLAIN ANALYZE select Name,Address from Student where Address!='Hyderabad';
                                                QUERY PLAN
------------------------------------------------------------------------------------------------------
 Seq Scan on student  (cost=0.00..16.00 rows=478 width=64) (actual time=0.027..0.031 rows=4 loops=1)
   Filter: ((address)::text <> 'Hyderabad'::text)
   Rows Removed by Filter: 2
 Planning Time: 0.577 ms
 Execution Time: 0.105 ms
(5 rows)

hostel_management=# _
```